

Bioprocess Engineering

Sem 4

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[Page 1]

MICROBIAL GROWTH AND

PRODUCT FORMATION KINETICS

Subject :- Bioprocess Principle

[Page 2]

1 Model classification- Model Formulation-

2 Unstructured Models

3 Phases of batch growth cycle

4 Monod Models

5 Multiple substrate models and model Inhibition,

6 Models of growth and non-growth product inhibition

7 Models for the growth of fungi, Plant cell and Animal cells,

8 Structured models

9 Models of metabolites and growth-compartmental Models

10 Models of product formation.

2nd Edition, Biochemical Engineering

By Douglas S. Clark, Harvey W.

Blanch

Biochemical Engineering Fundamentals

By Bailey and Ollis

Page no 383 - 438

[Page 4]

Mathematical Modelling

[Page 5]

Modelling is an iterative process

[Page 6]

Why do Mathematical Modelling?

- Mathematical modeling has always been an important activity in science and engineering.
- The formulation of qualitative questions about an observed phenomenon as mathematical problems was the motivation for and an integral part of the development of mathematics from the very beginning.
- Computational power is particularly important in modeling chemical engineering systems, as the physical and chemical laws governing these processes are complex.
- Besides heat, mass, and momentum transfer, these processes may also

include chemical reactions, reaction heat, adsorption, desorption, phase transition, multiphase flow, etc.

- This makes modeling challenging but also necessary to understand complex interactions.

[Page 7]

Why do Mathematical Modelling?

- In a chemical engineering context, mathematical modeling is a prerequisite for:
 - Design and scale-up;
 - Process control;
 - Optimization;
 - Mechanistic understanding;
 - Evaluation/planning of experiments;
 - Trouble shooting and diagnostics;
 - Determining quantities that cannot be measured directly;
 - Simulation instead of costly experiments in the development lab;
 - Feasibility studies to determine potential before building prototype equipment or devices.

[Page 8]

Modelling Procedure

- Each step in the modeling process requires an understanding of a variety of concepts and techniques blended with a combination of critical and creative thinking, intuition and foresight, and decision making.

Different steps in the Model Development

[Page 9]

- Step 1: Problem definition:
 - This involves stating clear goals for the modeling, including the various elements that pertain to the problem and its solution. (what hypothesis and which assumptions to use)
- Step 2: Formulation of conceptual model:
 - Collect data and experience about the subject to be modeled.
 - The main challenges are in identifying the underlying mechanisms and governing physical/chemical principles of the problem.
- Step 3: Formulation of mathematical model:

- Each important quantity is represented by a suitable mathematical entity, e.g. a variable, a function, a graph, etc.
- What are the variables (dependent, independent, parameters)?
- What are the constraints?
- What boundary conditions?
- What initial conditions?
- Step 4: Solution of the mathematical problem:
 - Check the validity of individual mathematical relationships, and whether the relationships are mutually consistent.
 - Verify the mathematical solution, i.e. ensure that you have solved the equations correctly. This step involves checking your solution against previously known results (analytical/numerical), simplified limiting cases, etc.

[Page 10]

- Step 5: Estimation of parameters:
 - The parameters of the system must be evaluated and the appropriate values must be used in the model. Some parameters can be obtained independently of the mathematical model. (e.g. gravitation constant, solubility data)
 - However, it is usually not possible to evaluate all the parameters from specific experiments, and many of them have to be estimated by taking results from the whole (or a similar system), and then using parameter-fitting techniques to determine which set of parameter values makes the model best fit the experimental results.
 - For example, a complex reaction may involve ten or more kinetic constants.
 - These constants can be estimated by fitting a model to results from a laboratory reactor.
- Step 6: Evaluation/validation :
 - A key step in mathematical modeling is experimental validation.
 - Ideally the validation should be made using independent experimental results, i.e. not the same set as used for parameter estimation.
 - During the validation procedure it may happen that the model still has some deficiencies.
 - In that case, we have to “iterate” the model and eventually modify it.
 - Certain characteristics that models which have a bearing on the question of how good they are: Accuracy; Descriptive realism; Precision; Robustness; Generality; Fruitfulness.
- Step 7: Interpretation/application:
 - The validated model is then ready to be used for one or several purposes as described earlier, e.g. to enhance our understanding, make predictions, and give information about how to control the process.

[Page 11]

Mathematical Model:

Use of mathematical model

Research and

Development Plant OperationDesign

- Determining chemical kinetics mechanisms and parameters from pilot-scale or lab-scale
- Aiding in scale-up condition
- Exploring the sizing and arrangement of processing equipment's
- Simulating start-up, shut down, and emergency situations and procedures
- Troubleshooting control and processing problems
- Studying the effect of and requirements for expansion projects

Principle of formulation:

Basis Assumption

Mathematical

consistency of

model

Solution of the

model equations

Verificati

on

[Page 12]

Classification of Mathematical

Models

- Mathematical models are classified by
 - Grouping into opposite pairs;
 - Mathematical complexity;
 - Degree of resolution.

[Page 13]

Grouping of models into opposite

pairs

- Linear versus nonlinear;
- Steady state versus non-steady state;
- Lumped parameter versus distributed parameter;
- Continuous versus discrete variables;
- Deterministic versus stochastic;
- Interpolation versus extrapolation;
- Mechanistic versus empirical;
- Coupled versus not coupled.

[Page 14]

Classification of modelling

techniques:

Modelling

techniques

Fuzzy model (Monad
model)

Discrete event model

Stochastic

model/Probabilistic

Population balance
model

Unsteady state model

Steady state model

Partial differentiation
model

Distributed parameter
model

Lumped parameters
model

[Page 15]

Linear versus nonlinear Steady state versus non-steady state

[Page 16]

Interpolation versus extrapolation

Danger of extrapolation. Yield of a
chemical reactor versus time

[Page 17]

Mechanistic model Empirical model

S

C B

A

S

$$f(x) = s$$

Where 's' is over all
effect

$$f(x) = A+B+C$$

Where f(x) is the sum of individual effects

of A,B and C
Example : Growth of micro organism in sludge

Growth of micro organism is given

by $f(x) = s$

Where 's' is over all nutrient effect

Growth of micro organism is given by

$$f(x) = \text{Nitrogen} + \text{Carbon} + \text{Nutrients}$$

Where f(x) is the sum of individual effects

of nitrogen, carbon and nutrients

Mechanistic model of microbial growth in sludge Empirical model of microbial growth in sludge

[Page 18]

Mechanistic model:

Mechanistic

model

Characteristics

Understand underlying concept Explain individual components

Demerits Merits

Accurate representation of a
project

Understanding a problem

Analyse weakness of individual
point

Help in redesigning components

Testable and verifiable

Precision in data may be
compromised

Time & cost ineffective

1

1

Figure 4 : Overview of mechanistic model

[Page 19]

Classification based on Mathematical complexity

[Page 20]

Classification of models according to scale

Macroscopic (Entire System) Microscopic (Elemental)

[Page 21]

21

Bioprocess modeling

In order to improve process understanding or performance, different automatic tools can be developed: simulators, which are able to reproduce system behaviors, software sensors which allow to obtain an estimation of an unmeasured signal or controllers to maintain optimal conditions. All these tools rely on representation of the biosystem, and a mathematical model.

[Page 22]

22

Characteristics of bioprocess models

There are two new characteristics for bioprocess models. First, a model can be structured or unstructured depending on whether it describes intracellular characteristics of the cell (metabolic reactions, cellular processes etc.) or (considers the cell like an entity without internal structure).

The next class of models are segregated or unsegregated depending on whether it considers the heterogeneity of the cellular population, or the position in the cell cycle..

[Page 23]

Such reaction rates vary with time and are usually influenced by many physicochemical and biological environmental factors like substrate, biomass and product concentrations as well as pH, temperature, dissolved oxygen concentration or various microbial growth inhibitors.

[Page 24]

Malthus law

- The simplest cell growth model is the Malthusian model commonly known as exponential law.
- overall rate of biomass formation is proportional to the biomass itself.

For a batch system,

- where x is the mass of cells per unit volume, μ is the proportionality constant known as 'specific growth rate' (h^{-1}) and t is the time (h).
- Here as the time increases, cell mass will also increase.
- $X = X_0 e^{\mu t}$
- The demerit of this model is that it ignores the substrate needed for its growth and that the resources are constrained.

$\frac{dx}{dt}$

$\frac{dx}{dt} = \mu x$

[Page 25]

Quantifying Growth Kinetics

- Structured vs Unstructured -
 - S - model divides the cell mass into components
 - U - assumes fixed cell composition - exponential growth phase of batch or in continuous
- Segregated vs Nonsegregated
 - S - assumes different types of cells exist
 - N - all cells are the same type

[Page 26]

Unstructured, Nonsegregated Models (Monod)

- Assumptions
 - One limiting substrate
 - Semi empirical relationship
 - Single enzyme system with Michaelis-Menten kinetics is responsible for the uptake of substrate
 - Amount of enzyme is sufficiently low to be growth limiting
 - Low population density
- Most commonly used expression for growth

SK

S

S

m

μ μ_m

μ μ_m - maximum growth rate when $S \gg$

K_s

- K_s - saturation constant - concentration of the rate-limiting substrate when the specific rate of growth is equal to one half of the maximum.
- $K_s = S$ when $\mu = 1/2\mu_m$
- In general $\mu = \mu_m$ for $S \gg K_s$ and for $S \ll K_s$

[Page 27]

Cell Growth - Monod

0

0.1

0.2
 0.3
 0.4
 0.5
 0.6
 0.7
 0 5 10 15 20 25
 Time (hr)
 Cell Concentration (g/L)
 Does not account
 for death phase
 Does not account
 for lag phase

[Page 28]

Unstructured, Nonsegregated Models

- Assuming one limiting substrate
 - Blackman
 - Tessier
 - Moser
 - Contois
- Multiple Substrates are growth limiting
 - Usually do not use unstructured, nonsegregated model

[Page 29]

Other unstructured, nonsegregated models (assuming one limiting substrate):

Blackman equation: $\mu = \mu_m$ if $CS \geq 2KS$

$\mu =$
 $\mu_m CS$

$2 KS$

if $CS < 2KS$

Tessier equation: $\mu = \mu_m [1 - \exp(-KCS)]$

Moser equation: $\mu =$

$\mu_m CS$

n

$KS + CS$

n

Contois equation: $\mu =$

$\mu_m CS$

$KSX CX + CS$

[Page 30]

M.PANDIMADEVI 30

Blackman equation:

0
0.2
0.4
0.6
0.8
1
0 2 4 6 8 10
Cs (g/L)
 $\mu(\text{per h}) \mu_m = 0.9 \text{ per h}$
 $K_s = 0.7 \text{ g/L}$
 $\mu = \mu_m$
 $\mu =$
 $\mu_m \frac{C_s}{K_s + C_s}$
if $C_s \geq 2 K_s$
if $C_s < 2 K_s$

This often fits the data better than the Monod model, but the discontinuity can be a problem.

[Page 31]

Tessier equation:

0
0.2
0.4
0.6
0.8
1
0 2 4 6 8 10
Cs (g/L)
 $\mu(\text{per h}) \mu_m = 0.9 \text{ per h}$
 $K = 0.7 \text{ g/L}$
 $\mu = \mu_m [1 - \exp(-K C_s)]$

[Page 32]

Moser equation:

0
0.2
0.4
0.6
0.8
1
0 2 4 6 8 10
Cs (g/L)

Monod

$$n = 0.25$$

$$n = 0.5$$

$$n = 0.75$$

$$\mu(\text{per h})$$

$$\mu =$$

$$\mu_m \frac{C_S^n}{K_S + C_S^n}$$

$$n$$

$$K_S + C_S^n$$

$$n$$

$$\mu_m = 0.9 \text{ per h}$$

$$K_S = 0.7 \text{ g/L}$$

When $n = 1$, Moser equation describes Monod model.

[Page 33]

Contois equation:

Saturation constant ($K_{SX} C_X$) is proportional to cell

concentration $\mu =$

$$\mu_{\max} \frac{C_S}{K_{SX} C_X + C_S}$$

$$K_{SX} C_X + C_S$$

Verlhurst model

Verlhurst kinetic model depends on the cell concentration. This model has two kinetic constants of maximum specific growth rate ($\mu_{\max}$) and maximum cell concentration ($x_{\max}$). Verlhurst model is expressed by the following equation

$$\mu = \mu_{\max} - (\mu_{\max} / X_{\max}) * X$$

Andrew model

Andrew model is proposed by the following equation for the growth dependent which incorporate substrate inhibition.

$$\mu = \mu_{\max} / [(1 - k_s/s)(1 + s/K_{is})]$$

[Page 34]

Extended Monod model:

$$\theta$$

$$0.2$$

$$0.4$$

$$0.6$$

$$0.8$$

$$1$$

$$0 \ 2 \ 4 \ 6 \ 8 \ 10$$

$$C_S \text{ (g/L)}$$

$$\mu(\text{per h}) \quad \mu_m = 0.9 \text{ per h}$$

$$K_S = 0.7 \text{ g/L}$$

$$C_{S,\min} = 0.5 \text{ g/L}$$

$$\mu =$$

$$\mu_m \left(\frac{C_S - C_{S,\min}}{C_S} \right)$$

$K_S + C_S - C_{S,min}$

Extended Monod model includes a $C_{S,min}$ term,
which denotes the minimal substrate
concentration needed for cell growth.

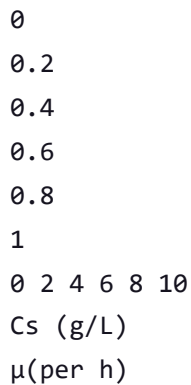
[Page 35]

M.PANDIMADEVI 35

Monod model does not model substrate inhibition.

Substrate inhibition means increasing substrate concentration beyond certain value reduces
the cell growth rate.

Monod model modified for substrate inhibition:



[Page 36]

$$\mu = \frac{\mu_m C_S}{K_S + C_S + \frac{C_S^2}{K_I}}$$

where K_I is the substrate inhibition constant.

Monod model modified for cell growth with
noncompetitive substrate inhibition:

$$\mu = \frac{\mu_m (1 + K_S/C_S)(1 + C_S/K_I)}{(1 + K_S/C_S + C_S^2/(K_I C_S))}$$

[Page 37]

$$\mu = \frac{\mu_m C_S}{K_S + C_S + \frac{C_S^2}{K_I}}$$

where K_I is the substrate inhibition constant.

Monod model modified for cell growth with

noncompetitive substrate inhibition:

$$\mu = \frac{\mu_m}{(1 + K_S/C_S)(1 + C_S/K_I)}$$

If $K_I \gg K_S$ then

$$\mu = \frac{\mu_m C_S}{K_S + C_S + C_S^2/K_I}$$

[Page 38]

where K_I is the substrate inhibition constant.

Monod model modified for cell growth with competitive substrate inhibition:

$$\mu = \frac{\mu_m C_S}{K_S(1 + C_S/K_I) + C_S}$$

Monod model for two limiting substrates

$$\mu = \frac{C_{S1}}{K_{S1} + C_{S1}} \frac{C_{S2}}{K_{S2} + C_{S2}}$$

[Page 39]

Monod model modified for cell growth

with product inhibition:

Monod model does not model product inhibition (where increasing product concentration beyond certain value reduces the cell growth rate)

where C_p is the product concentration and K_p is a product inhibition constant.

For competitive product inhibition:

For non-competitive product inhibition:

$$\mu = \frac{\mu_m}{(1 + K_S/C_S)(1 + C_p/K_p)}$$
$$\mu = \frac{\mu_m C_S}{K_S(1 + C_p/K_p) + C_S}$$

[Page 40]

Monod model modified for cell growth with

product inhibition:

where C_{pm} is the product concentration at which growth stops.

Ethanol fermentation from glucose by yeasts is an example of non-competitive product

inhibition. Ethanol is an inhibitor at concentrations above nearly 5% (v/v). Rate expressions specifically for ethanol inhibition are the following:

$$\mu = \frac{\mu_m CS}{(KS + CS)(1 + C_p/K_p)}$$

$$\mu = \frac{\mu_m CS}{(KS + CS) \exp(-C_p/K_p)}$$

[Page 41]

Monod model modified for cell growth with toxic compound inhibition:

where CI is the product concentration and KI is a constant to be determined.

For competitive toxic compound inhibition:

For non-competitive toxic compound inhibition:

$$\mu = \frac{\mu_m}{(1 + KS/CS)(1 + CI/KI)}$$

$$\mu = \frac{\mu_m CS}{KS(1 + CI/KI) + CS}$$

[Page 42]

Monod model extended to include cell death kinetics:

where kd is the specific death rate (per time).

$$\mu = \frac{\mu_m CS}{KS + CS} - k_d$$

[Page 43]

Unstructured Batch Growth

Models• A brief discussion on modelling of bacterial growth has been discussed in the beginning of the course using the simplest approach of the form,

- Where from malthus law equation $f(x)=\mu x$
- μ =constant , describing the microbial growth rate
- Verlhurst in 1844 and pearl and reed 1920 put forwarded the theory which included inhibiting factor to population growth assuming that inhibition is proportional to square of cellular biomass

dc

$$C(0)=0$$

Another class of stationary approach a stationary population level can be formulated based on limiting nutrient exhaustion. Assuming a constant overall yield factor and

Such as monod kinetics, nutrient and cell material balance can be combined into a single equation of the form . A drawback of the logistic equation is its failure to predict a phase of decline after the stationary population has exhausted all available resources.

This feature is found in one model developed by Volterra
Physically we may interpret such a term as a crude recognition that the history $k(t,r)$ of a population influence its growth rate.

[Page 46]

$$\frac{dx}{dt} = kx \left(1 - \frac{x}{K_0} \right)$$

Specifically, let us suppose that k is a constant equal to k_0 and that the history.

Then the population size is described by

K_0 the sign is taken as negative for an inhibitor and positive for a compound which promotes growth.

K_0 is negative the population size decline after passing through the maximum.

[Page 47]

Product Formation Kinetics

Unstructured model

Simplest type of product formation kinetics arise when there is a simple stoichiometric connection between product formation and substrate utilization or cell growth. The rate of product formation can be written as

$$r_p = f_{SP} \mu$$

It can also be expressed as

$$r_p = f_{XP} \mu$$

For anaerobic fermentation such as *Lactobacillus delbrukii*, the kinetics of product formation is famously expressed by the Leudeking-Piret kinetics given by Such a form is normally used in establishing product formation data. The first term in the above equation refers to the energy used for growth and the second term refers to the energy used for maintenance.

x_r
 $i_p = f_f \frac{1}{1 + \frac{x_r}{K_x}}$

[Page 48]

Consider a batch fermentation model given by a logistic model below

$\frac{dx}{dt} = \mu_{max} x \left(1 - \frac{x}{K_x} \right)$

and product formation is given by Leudeking-Piret kinetics of the form

$\frac{dx}{dt}$

$\frac{dx}{dt}$

$\frac{dx}{dt}$

$\frac{dp}{dt} = \frac{1}{Y_{p/x}} \frac{dx}{dt}$

And substrate utilization is given by

$\frac{dx}{dt}$

$\frac{dp}{dt}$

Y_{dt}

$\frac{dx}{dt}$

Y_{dt}

$\frac{ds}{dt}$

$\frac{ds}{dt}$

$\frac{ds}{dt} = -\frac{1}{Y_{s/x}} \frac{dx}{dt}$

For substrate consumption,

Substitute $\frac{dp}{dt}$ into $\frac{ds}{dt}$ gives

$\frac{dx}{dt} = \mu_{max} x \left(1 - \frac{x}{K_x} \right)$

$\frac{dx}{dt}$

Y_{Ydt}

$\frac{ds}{dt}$

$\frac{ds}{dt}$

$\frac{ds}{dt} = -\frac{1}{Y_{s/x}} \frac{dx}{dt}$

This equation reduce into

$\frac{dx}{dt}$

$\frac{dx}{dt}$

$\frac{dx}{dt}$

$\frac{ds}{dt} = -\frac{1}{Y_{s/x}} \frac{dx}{dt}$

[Page 49]

$\frac{ds}{dt}$

$\frac{ds}{dt} = -\frac{1}{Y_{s/x}} \frac{dx}{dt}$

$\frac{ds}{dt} = -\frac{1}{Y_{s/x}} \frac{dx}{dt}$

Were as

$\frac{dx}{dt} = kspY \frac{1}{1 + \frac{K}{x}}$

and

Integrating dx/dt gives a Riccati type equation of the form

$\frac{dx}{dt} = \frac{1}{1 + \frac{K}{x}}$

$\frac{dx}{dt}$

kx

kx

e

ex

x

x

$\frac{dx}{dt} = \frac{1}{1 + \frac{K}{x}}$

By rearrange this equation

$\frac{dx}{dt} = \frac{1}{1 + \frac{K}{x}}$

$\frac{dx}{dt}$

$\frac{dx}{dt}$

$\frac{dx}{dt}$

$\frac{dx}{dt}$

$\frac{dx}{dt} = \frac{1}{1 + \frac{K}{x}}$

$\frac{dx}{dt}$

$\frac{dx}{dt}$

$\frac{dx}{dt}$

$\frac{dx}{dt}$

$\frac{dx}{dt}$

$\frac{dx}{dt}$

$\frac{dx}{dt}$

$\frac{dx}{dt}$

$\frac{dx}{dt}$

$\frac{dx}{dt}$

$\frac{dx}{dt}$

$\frac{dx}{dt}$

$\frac{dx}{dt}$

$\ln(1 + \frac{K}{x})$

$\ln(1 + \frac{K}{x})$

$\ln(1 + \frac{K}{x})$

$\ln(1 + \frac{K}{x})$

$\ln(1 + \frac{K}{x})$

$\ln(1 + \frac{K}{x})$

$\ln(1 + \frac{K}{x})$

$\ln(1 + \frac{K}{x})$

$\ln(1 + \frac{K}{x})$

$\ln(1 + \frac{K}{x})$

$\ln(1 + \frac{K}{x})$

$\ln(1 + \frac{K}{x})$

Where β is substituted for stationary cell, x_s

Plot a graph using above equation to find K as slope , and $-\ln(X_s/X_0-1)$ as a intercept in y- axis

[Page 50]

Structured kinetics Model

j
 c
 f_R
 c
 j_r
 c
 $c x_r v x c v d t$
 d
 j
 $\frac{d}{dt}$

111 $\frac{d}{dt}$

$\frac{d}{dt}$
 $\frac{d}{dt}$
 $\frac{d}{dt}$

Unstructured model of the above type describe only the quantity of the biological phase. Such models do not recognize nor represent the composition or what we might call the quality of the biophase. In situation in which the cell population composition changes significantly and in which these composition changes influence kinetics, structured model should be used. Since writing the material balance equation for every component of cell is not possible

The biophase variable employed in structured model are typically the mass x_j , molar concentration c_j per unit volume of biophase, for a well mixed reactor the material balance on component j gives

Where ρ_c = mass density of cell (cell mass/ unit volume cells)

r_{fi} = molar rate of formation of component J

Φ_x = mas of the cell added to the recator by flow. Time -1

V_R = culture volume

X = cell mass concentration (cell mass/unit volume culture)

C_j = moles J /Unit volume cells

[Page 51]

R
 x
 $j j f$
 j
 $x V c d t$
 $d x$
 $x c r d t$
 $d c$
 j

???

?

??

?

???

jf

j

crdt

dc

j

???

It is assumed that any cells added or removed from the reactor have the Same state as the cell population in the reactor. If this condition is not met, The balance equation must be modified.

Assuming the density of the cells pc and volume of culture VR are time invariant may be rewritten in the following form after carrying out the indicated differentiation

For a batch reactor, Φx is zero and the quantity in parenthesis on the right hand is simply the specific growth rate μ , the equ

The Physical interpretation of $r f j$, the $-\mu c j$ term represent reduction in concentration caused by dilution from growth of the cell population.

[Page 52]

For product formation , using the following eq

$\frac{dx}{dt}$

$\frac{dx}{dt}$

$\frac{dx}{dt}$

$\frac{dp}{dt}$

With riccate equation and upon integration gives

The constant β is found at the stationary phase of batch culture

$\beta = \frac{1}{\mu_{max}} \ln \left(\frac{K_s + x_{max}}{K_s + x} \right)$

β

$\beta = \frac{1}{\mu_{max}} \ln \left(\frac{K_s + x_{max}}{K_s + x} \right)$

β

β

$\beta = \frac{1}{\mu_{max}} \ln \left(\frac{K_s + x_{max}}{K_s + x} \right)$

β

$\beta = \frac{1}{\mu_{max}} \ln \left(\frac{K_s + x_{max}}{K_s + x} \right)$

β

stationary

$\frac{dp}{dt}$

??

$\beta = \frac{1}{\mu_{max}} \ln \left(\frac{K_s + x_{max}}{K_s + x} \right)$

β

$\beta = \frac{1}{\mu_{max}} \ln \left(\frac{K_s + x_{max}}{K_s + x} \right)$

β

k
xptp kt
s
s $\frac{dX}{dt}$ $\frac{dP}{dt}$

And μ is determined by a plot of,

[Page 53]

The aerobic bioprocess modeling is an useful tool to accomplish several important tasks

- (a) it can be the basis for adequate optimization and control technique applications;
- (b) it can provide the necessary information about the features of the chosen bioprocessing system;
- (c) it synthesizes the characteristics of the specified living cells' evolution and hence, it is the best technique to predict the process efficiency.

[Page 54]

- The unstructured global models are in use nowadays as the main tool for both the bioprocess modeling, but also for being applied in overall computer control.
- Their limit is they are a simplified representation of the bioprocess behavior: conforming to this concept the bioprocess evolution depends directly and only on the macroscopic variables representing the working conditions in the bioreactor.
- Therefore the unstructured models are essentially kinetic equations that describe the variation of substrate or product concentrations and of a unique biological state variable-the cell concentration, and can also express the influences of some important process variables (pH, pO_2 , temperature, and others), and only sometimes they are balance equations.

[Page 55]

- the structured models - the biotic phase is not any more viewed as a homogenous component, but they provide information about the physiological state of the cells, their composition and regulatory adaptation to the environment.
- Conforming to this concept the cell mass is structured in several intracellular compounds and functional groups, which are connected to each other and to the environment by fluxes of material and information.
- The structured models can be:
 - multi compartment models,
 - genetically structured models, and

- biochemical structured models.

[Page 56]

Structured product formation kinetic modeling

- Only little development for the structured kinetics of product formation
- Models are usually developed based on metabolite production
- Most are based on antibiotic fermentation such as *Claviceps purpurea*
- The model is divided into four steps, the cellular growth, the extracellular and intracellular phosphate levels and the product formation.

For cellular growth

$\frac{dx}{dt} = \mu x$

21

$\frac{dx}{dt} = \mu x$

$\frac{dx}{dt} = \mu x$

For extra cellular Phosphates

$\frac{dx}{dt} = \mu x$

2

2

3 $\frac{dx}{dt} = \mu x$

$\frac{dx}{dt} = \mu x$

$\frac{dx}{dt} = \mu x$

$\frac{dx}{dt} = \mu x$

[Page 57]

For extra cellular phosphate

$\frac{dx}{dt} = \mu x$

1

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$\frac{dx}{dt} = \mu x$

$\frac{dx}{dt} = \mu x$

$\frac{dx}{dt} = \mu x$

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For Product formation

2

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4

$\frac{dx}{dt} = \mu x$

$\frac{dx}{dt} = \mu x$

$\frac{dx}{dt} = \mu x$

$\frac{dx}{dt} = \mu x$

[Page 58]

Compartment model

- ☐ It is a simplest structured model
- ☐ compartmentalization of components/section into small sizes
- ☐ synthetic components such as RNA and precursors
- ☐ structural components such as DNA and proteins
- ☐ This model could also be defined as the assimilatory component and a synthetic component.
- ☐ class of transport and reaction occurring in a cell population include
- ☐ molecular collision
- ☐ chemical reactions
- ☐ diffusion
- ☐ RNA turnover
- ☐ protein synthesis
- ☐ increase in cell number
- ☐ completion of batch process
- ☐ spontaneous mutation

The Process as the relaxation time after a perturbation (it's a time to reach steady state)

[Page 59]

- This relaxation time range from fraction of a second to hour.
- what is important in modeling, from a dynamic view point is the relationship b/w the time scale of change in cells environment τ_E and the spectrum of relaxation time of cellular processes.
- Those cellular process which respond to fast environmental changes is referred as quasi steady state (small relaxation time/ τ_E ratio)
 - for large relaxation time compared to τ_E , cellular system may be assumed to be frozen at the initial state (example: significant genetic changes are not usually expected during a single batch cultivation)

[Page 60]

- Willams proposed a simple two compartment model with assumption such as

Component

1

Component

2

- ☐ The synthetic portion is produced by uptake of external nutrient S with yield coefficient Y . The rate of cell synthetic component formation is first order in total cell density x and nutrient mass concentration.
- ☐ The structural and genetics cell component is produced from component 1 at a rate proportional to p_1 and p_2
- ☐ doubling of component 2 is necessary and sufficient for cell division and the cell

density is proportional to the density of component 2 in the culture

□Biomass comprised entirely of components 1 and 2

s Rate constants

[Page 61]

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dx

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ds

Where

sk1□□

And

constc □□□ □□□ 21

Eq 1

[Page 62]

)1() (2221

2 ffksfkdt

df

cc □□□□ □

eq 1 can be rewritten has

Where f2 is

c□

□2 This fraction of cell mass comprised of component 2

Such a model has been used to simulate batch growth using a stationary, nutrient-exhausted inoculum that corresponds to $p_1 = 0$.

Simulation results in several frequently observed features of batch microbial

cultures, include

- Existence of lag phase ,cell size increases.
- Exponential growth phase, cell size is maximum.
- Change in composition of cells during growth cycle, since these changes are evident during exponential phase.
- Stationary phase with relatively small cells.

[Page 63]

Other unstructured, nonsegregated models

(assuming one limiting substrate):

Luedeking-Piret model:

$$r_P = \alpha r_X + \beta C_X$$

Used for lactic acid formation by *Lactobacillus debruickii* where production of lactic acid was found to occur semi-independently of cell growth.

[Page 64]

Student Learning Objective

(SLO)-7

Apply in monod Inibition kinetics on

Microbial Batch system

[Page 65]

Inhibition Models

(Very similar to enzyme models)

- Substrate Inhibition
 - High substrate concentration inhibits growth
 - If a single-substrate enzyme catalyzed reaction is the rate-limiting step then inhibition of enzyme activity results in inhibition of microbial growth.

Noncompetitive Substrate Competitive Substrate

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[Page 66]

Inhibition Models - cont.

- Product Inhibition

- High concentrations of product can be inhibitory
- Underlying mechanism of product inhibition is unknown
- Approximated as exponential or linear decay functions

Noncompetitive Product Competitive Product

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[Page 67]

Inhibition Models - cont.

- Inhibition by Toxic Compounds

- inhibition of growth is analogous to enzyme inhibition

Noncompetitive Competitive

Uncompetitive Cell Death

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[Page 68]

Batch Reactors

- Cell Growth
- Substrate Utilization
- Product (cometabolic contaminants use negative sign)

XSK

SXdt

dXr

S

m

X

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SXS
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 SK
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 dSr
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 dP
 Xq
 Ydt
 dP
 Xq
 S
 m
 p
 ggXpp
 1
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[Page 69]
 Logistic Equation
 • Batch Growth Equation
 • Combines Batch growth,
 Monod and Yield
 Coefficients
 • No maintenance
 0
 0.1
 0.2
 0.3
 0.4

0.5
 0.6
 0.7
 0 5 10 15 20
 Time (hr)
 Cell Concentration (g/L)

[Page 70]

Rate Expression for Growth

(1)

Yield Expression

(2)

Substituting into Eq. 1 for S from Eq 2:

Integrating

Logistic Equation

XSK

SXdt

dXr

S

m

X

$\frac{dX}{dt} = \mu_{max} X \left(\frac{S}{K_S + S} \right) \left(\frac{X_{max} - X}{X_{max}} \right)$

$\frac{dX}{dt} = \mu_{max} X \left(\frac{S}{K_S + S} \right) \left(\frac{X_{max} - X}{X_{max}} \right)$

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Describes

Sigmoidal shape

batch growth curve

[Page 71]

Unstructured, Nonsegregated Models

- Disadvantage of Unstructured, Nonsegregated Models
 - No attempt to utilize or recognize knowledge about cellular metabolism and regulation
 - no description on lag phase
 - give no insight to the variables that influence growth
 - assume a black box
 - assume dynamic response of a cell is dominated by an internal process with a time delay on the order of the response time
 - most processes are assumed to be too fast (pseudo r/n) or too slow to influence the observed response.

[Page 72]

Filamentous Organisms

- Types of Organisms
 - mold
 - bacteria or yeast entrapped in a spherical gel particle
 - formation of microbial pellets in suspension
 - Model - no mass transfer limitations
 - R - radius of the cell floc or pellet or mold colony
- Then the growth of the biomass (M) can be written as

const k_dt

dR

d ??

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Rkdt

dR

Rdt

dM

p2

2

3

3

4 RM 222

[Page 73]

Filamentous Organisms - cont.

- Where as:
- Integrating the equation:
- M_0 is usually very small yield then
- M will be cubic dependence Model is supported by experimental data 1.

33

3/1

0

33 2

2

22

2

222

2

22

2

2 22 ttMM 22

3tM 2

Mdt

dM 32

22

3/1)36(222 pk2

[Page 74]

REFERENCE

- Biochemical Engineering and fundamentals - James E. Bailey and David F. Ollis
- Bioprocess Engineering and Basic Concept- Micheal

L. Shuler and Fikret Kargi, page:176-183.

[Page 83]

Models of growth associated and non growth associated product formation

Growth growth associated product formation

- End products of energy and carbon metabolism are primary metabolism
- Examples; ethanol production by anaerobic fermentation glucose by yeast
- Production of gluconic acid from glucose by gluconobactor
- These products are referred to as growth associated products as their rate of production parallels the growth of the cell population

Non growth associated product formation

- Products such as vitamins and antibiotics are produced in a batch culture at the end of exponential phase
- Such secondary metabolites are termed as Non growth associated products
- Their kinetics does not depend on the rate of growth of the culture

[Page 84]

Partially growth associated product formation

- Product formation lies between the growth associated product formation and Non growth associated product formation
- Examples: Amino acids, lactic acid, intermediates from citric acid cycle (citric acid), extracellular polysaccharides (Xanthan and pullulan) and solvents such as acetone and butanol

[Page 85]

Luedeking pirt Model

[Page 86]

Growth associated kinetics

[Page 88]

Non - Growth Associated Kinetics :The penicillin fermentation

- Penicillin production can be considered to be non growth associated as first approximation
- The production of penicillin occurs after the phase of rapid growth of the fungus *Penicillium chrysogenum*. This second phase is referred to as idiophase
- At the end of growth phase, two key enzymes are found to increase in activity
 1. phenyl acetic acid- which form a side chain of penicillin
 2. penicillin acyl-transferase which attaches activated phenylacetate to 6 aminopenicilanic acid, which is the penicillin nucleus
- Penicillin production is subjected to catabolite repression
- A slow supply of carbon source is necessary to ensure that intermediate metabolites

do not accumulate and reduce production by end product inhibition

[Page 89]

- After the period of exponential growth on lactose (0- 24 hours) glucose is slowly fed to the culture and penicillin production increases from 24 to 140 hours

- The non growth rate associated behavior at specific growth rates greater than 0.015 hour^{-1}

- Below this growth rate, growth - associated product formation is seen

Partial Growth and non growth associated kinetics. Lactic acid Production

- The kinetics of lactic acid production by *Lactobacillus* provide an example of product kinetics that are intermediate between growth and non growth associated.

- Lactic acid- anaerobic fermentation of sugars such as glucose and lactose

- It is mainly used as stearyl-2-lactolate

- Although the anaerobic conversion of glucose to lactic acid is a direct route for energy production, the production kinetics are not solely growth associated

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